G Pavan Kumar

Portfolio: pavan.com | GitHub: **github.com** | **LinkedIn: LinkedIn_Pavan**

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PROFESSIONAL SUMMARY

Final-year Computer Science Engineering student with strong command over Data Structures, Algorithms, and Object-Oriented Programming. Proficient in Java and Python, with hands-on experience building scalable backend systems and full-stack applications using Node.js, ReactJS, and SQL/NoSQL databases. Passionate about solving complex problems through code, writing clean and modular software, and working in collaborative, Agile environments.

EDUCATION

Bachelor of Engineering - Computer Science and Technology

Saveetha School of Engineering, SIMATS – Chennai, India **(Oct 2022 – Present)**

GPA: 8.6/10 | Expected Graduation: May 2026

Relevant Coursework: Data Structures, Algorithms, OS, ML, AI, DBMS, Distributed Systems, Networking

SKILLS

- **Programming Languages:** Python, C++, Java, JavaScript, SQL
- **Tools:** Git, GitHub, Postman, Jupyter, VS Code, Docker (basic), Linux/Unix, Chrome DevTools, pdb
- **Frameworks & Libraries**: ReactJS, Node.js, Flask, FastAPI, TensorFlow, Keras, OpenCV, Scikit-learn
- **Databases Systems:** MongoDB, MySQL, PostgreSQL
- **Cloud & DevOps:** AWS (basic), IBM Cloud, REST API integration, CI/CD (basic), Version Control
- **Testing & Debugging:** JUnit, PyTest, unit testing, logging

EXPERIENCE

Frontend Developer – Freelance **(Feb 2023 – Present)**

- Built and deployed 4+ responsive full-stack websites using ReactJS, HTML, CSS, JavaScript, and APIs
- Optimized load times by up to 40% and improved SEO rankings, boosting client traffic by an average of 25%.
- Delivered pixel-perfect UI/UX with smooth API integrations, increasing client satisfaction and repeat projects.
- Practiced Agile communication and on-time delivery for 10+ clients, maintaining a 100% on-time delivery rate.
- Managed code via GitHub and conducted client code walkthroughs.
- Worked with cross-functional teams including designers and backend developers for client delivery.

PROJECTS

- **Image Captioning Project:** Developing an advanced AI-driven image captioning system leveraging GPT-based models, surpassing traditional methods like ResNet-LSTM in generating more detailed, contextually relevant, and readable captions. **Tech:** Python, TensorFlow, GPT-based models, ResNetLSTM. **(2025)**

- **Laptop Sales and Service Project:** Developed an open-source platform for seamless laptop sales, service requests, and inventory management, enhancing customer experience and support efficiency.
  Tech: HTML5, CSS, React, JS, NodeJS, SQL, MongoDB. **(2024)**

- **Vehicle License Plate Detection System:** Built an automated license plate detection system using Convolutional Neural Networks (CNNs) and traditional machine learning algorithms for accurate vehicle identification.
  Tech: Python, OpenCV, CNN, SVM, Scikit-learn. **(2024)**
- **Real-Time Chat Application:** Built a secure, scalable real-time chat application using NodeJS and Socket.io, enabling instant communication and message storage in MongoDB.
  Tech: NodeJS, Socket.io, MongoDB. **(2024)**
- **Social Media Analytics Dashboard:** Created a dynamic dashboard for analyzing and visualizing social media engagement metrics from platforms like Twitter, Facebook, and Instagram.
  Tech: Python, Django, APIs. **(2024)**

HONORS AND AWARDS

- Top Performer – DevForge Coding Challenge, Chennai Institute of Technology-Earned "Top Performer" badge for full-stack development skills and problem-solving in backend infrastructure tasks. **(Oct 2024)**
- Winner – CodeX Hackathon, Saveetha School of Engineering-1st place for building a real-time license plate detection system using CNN and OpenCV. **(Nov 2023)**
- Top 5 – VIT Code Carnival, Vellore Institute of Technology-Recognized among top 5 teams for a social media analytics dashboard showcasing multi-platform engagement metrics. **(Nov 2024)**

PUBLICATIONS

- Navigating Urban Dynamics for Evaluating the Impact of Deep Learning on Number Plate Recognition in Comparison of Convolutional Neural Networks vs Pixel-Based Approaches for Analyzing Efficiency Enhancement **(Nov 2023)**
- **Redefining Image Captioning:** GPT-Based Models Surpass Transformer Networks in Contextual Accuracy and Fluency. **Tech:** Flutter, Android, Firebase, TensorFlow, Python, Dart **(May 2025)**